

1. Explain the fundamental differences between traditional programming and Machine Learning!

Answer:

- **Traditional Programming:** In this approach, a programmer must explicitly write the rules, logic, and algorithms. The computer takes these predefined **rules** and the input **data** to process them and produce the **answers** (output). If the scenario changes, the programmer must manually update the code.
- **Machine Learning (ML):** In ML, the paradigm is reversed. The programmer provides the computer with historical **data** and the correct **answers** (labels/outcomes). The ML algorithm analyzes this to discover patterns and automatically generates the **rules** (the model). This model can then be used to evaluate new, unseen data to predict future answers without needing explicit rule programming.

2. Mention and explain the 3 types of Machine Learning along with examples of their application in web applications!

Answer:

1. **Supervised Learning:** The model is trained using labeled data (data paired with correct answers). It learns to map inputs to outputs.
 - *Web Application Example:* A spam filter in a webmail application. It is trained on thousands of emails labeled as "spam" or "not spam" to predict whether an incoming email belongs in the inbox or the spam folder. Another example is Sentiment Analysis on a product review website.
2. **Unsupervised Learning:** The model is given raw, unlabeled data and must find hidden structures, patterns, or groupings on its own.
 - *Web Application Example:* Customer segmentation in an e-commerce platform. The system groups users based on their browsing and purchasing behavior without prior labels, allowing the platform to generate personalized product recommendations (Recommendation Engines).
3. **Reinforcement Learning:** The model learns by interacting with an environment, operating on a trial-and-error basis. It receives rewards for correct actions and penalties for incorrect ones, aiming to maximize the total reward.
 - *Web Application Example:* An intelligent customer service Chatbot on a website. If it provides a helpful answer and the user gives a "thumbs up," it receives a reward, learning over time to provide more accurate and contextually appropriate responses.

3. Explain the 2 main architectures in integrating Machine Learning models into web applications! What are their advantages and disadvantages?

Answer:

1. Server-Side Architecture (API-Based Inference):

The ML model resides on a backend server (often using Python frameworks like Flask or FastAPI). The web frontend (client) sends user data via an API request to the server, which runs the prediction and sends the result back to the client.

- **Advantages:** Can handle massive, complex models; utilizes powerful server hardware (GPUs); keeps the model's intellectual property hidden from users; easy to update the model without touching the frontend code.
- **Disadvantages:** Requires constant internet connection; higher latency due to network round-trips; server hosting costs can be high with heavy traffic.

2. Client-Side Architecture (In-Browser Inference):

The ML model is downloaded and run directly inside the user's web browser using JavaScript libraries like TensorFlow.js.

- **Advantages:** Zero network latency during prediction (real-time interaction); high privacy since sensitive user data (like webcam feed or voice) never leaves the device; works offline once the model is cached.
- **Disadvantages:** Limited by the computing power of the user's device (can lag on older phones); models must be heavily compressed/small to avoid long webpage loading times; the model architecture is exposed to the public.

4. What is TensorFlow.js? Explain its advantages compared to running ML models on the server side!

Answer:

TensorFlow.js is an open-source library created by Google that allows developers to build, train, and run Machine Learning models directly in the browser using JavaScript (or in backend environments using Node.js).

Advantages over server-side models:

- **Zero Install / Easy Access:** Users don't need to install any complex drivers or software; they just open a web URL.
- **Data Privacy:** Because the inference happens directly on the client's machine, sensitive data (like facial recognition data from a webcam or medical document scans) does not need to be transmitted to a third-party server, greatly enhancing security and privacy.
- **Interactive Real-Time Performance:** Without the need to send data back and forth over a network (which causes lag), TensorFlow.js can process high-frame-rate webcam video or real-time audio smoothly, enabling highly interactive web experiences (e.g., augmented reality filters or gesture controls).
- **Cost Efficiency:** The computational burden is shifted from the developer's cloud servers to the user's device hardware (using WebGL for acceleration), drastically reducing server hosting and scaling costs.

5. Provide an idea for an innovative web application that utilizes the

integration of Natural Language Processing (NLP) or Computer Vision. Explain its main features!

Answer:

App Idea: "VirtuFit - The AI Web Wardrobe"

An e-commerce web application focused on fashion that integrates both Computer Vision and NLP to provide a highly personalized shopping experience.

Main Features:

1. Virtual Try-On (Computer Vision):

Using TensorFlow.js in the browser, the app asks permission to use the user's webcam. It utilizes pose-estimation (like MoveNet) to detect the user's body shape and movements in real-time. It then virtually overlays clothing items onto the user's body on the screen, moving realistically as the user turns around, allowing them to "try before they buy" directly from their browser without sending video data to a server.

2. "Style Guru" Assistant (Natural Language Processing):

Instead of manually filtering through hundreds of categories, the user can chat with an NLP-powered search assistant. The user can type or speak complex queries like: *"I am going to a semi-formal outdoor wedding next week in Bali, and I prefer dark colors."* The NLP engine analyzes the intent and extracts entities (Event: Wedding, Formality: Semi-formal, Setting: Outdoor/Tropical, Color: Dark). It then instantly queries the product database to curate and present an entire outfit recommendation that matches the specific context and weather of the request.